Aarav Verma

aarav.verma.eng0118@gmail.com | (360) 522-0227 | Engineering Portfolio | LinkedIn

EDUCATION

University of Illinois at Urbana-Champaign — Graduation: May 2029
Bachelor of Science, Aerospace Engineering, Grainger College of Engineering
Relevant Coursework: Linear Algebra, Differential Equations, Physics Electricity & Magnetism, CS124: Java, Aerospace Fundamentals

SKILLS

CAD & Rapid Prototyping: SolidWorks, Autodesk Fusion, Siemens NX, Onshape, 3D Printing (PLA, ABS), GD&T
Simulation & Analysis: ANSYS (FEA, Static Structural), MATLAB (PDE Toolbox), Structural Analysis, Mesh Convergence, Tolerance Programming & AI: Python (PyQt5, Matplotlib, NumPy), Java (OOP, Data Structures), React.js, Node.js, PostgreSQL, MATLAB, Git, Linux, LLM-Assisted Development

WORK EXPERIENCE

National Organization for Business and Engineering - Champaign, IL — Feb 2026-May 2026
Engineering Lead, Canary Drone Project

- Redesigned propeller from standard 2-blade to toroidal configuration in Fusion, reducing blade-tip vortex losses by enclosing the tip path and improving thrust efficiency by ~11% at 4-6k RPM while lowering noise signature below 60 dB at 1m hover
- Validated structural integrity of the toroidal propeller and motor mount assembly under 12N thrust loads using FEA in Fusion confirming a factor of safety above 2.0, and developed a fully cohesive BOM tracking 50+ components
- Designed a modular drone airframe in Fusion featuring a Jetson computer housing frame and foldable leg motor assembly with 4 snap-lock hinge joints, reducing field setup time and enabling tool-free reconfiguration between flight missions
- Preparing the Canary drone for the Cozad New Venture Challenge, competing against 300+ teams and 900+ students for $500,000 in funding by developing investor pitch materials, and refining the final flight ready prototype

Ghost Electric Motorcycles - Champaign, IL — Aug 2025-Jan 2026
Drivetrain Engineer

- Engineered a rear suspension swingarm in SolidWorks by reverse-engineering the 2010 Kawasaki ZX-6R, calculating anti-squat ratio and pivot geometry to replicate OEM kinematics, and validating stiffness, braking, and acceleration loads using FEA
- Designed a load-bearing drivetrain frame in SolidWorks to house and align motor, top rod, and swingarm subassemblies ensuring precise mounting interfaces to $\pm$0.05 mm tolerances
- Designed and prototyped a load-bearing drivetrain top rod in SolidWorks, producing manufacturing-ready drawings with GD&T-defined tolerances and validating fit through iterative 3D-printed prototype testing
- Performed structural analysis on the drivetrain assembly in ANSYS FEA and MATLAB, validating stiffness and load capacity under peak torque deliver, chain tension, and rider-weight loading conditions across breaking, cornering, and max acceleration

PROJECTS

Parametric Airfoil Structural Analysis - Olympia WA — Aug 2026
MATLAB (PDE Toolbox), Fusion 360

- Generated NACA 0012 airfoil geometry in MATLAB, modeled the solid in Fusion 360, and imported STEP geometry into MATLAB for structural FEA under directional surface loads
- Diagnosed a stress singularity through a mesh convergence study where peak stress diverged with refinement, then added a CAD fillet that brought successive mesh results within 2% of each other

Cantilever Bracket FEA - Olympia WA — Aug 2026
MATLAB, Fusion 360

- Predicted tip deflection and bending stress by hand from beam theory, then caught a 10x error in the FEA load application by comparing against those calculations and querying the true load-face area from the solved mesh
- Reconciled corrected FEA output to within 10% of hand calculations, attributing the remaining gap to bend-fillet compliance and mounting-hole stress concentrations the beam model omits

Thrust Vector Control (TVC) Rocket Flight Simulator - Champaign, IL — Nov 2025-Dec 2025
Python, PyQt5, Matplotlib

- Built a 6-DOF rocket flight simulator in Python to develop a foundational understanding of TVC systems incorporating quaternion attitude dynamics, RK4 numerical integration, mass-varying propulsion, and gimbal-to-torque TVC modeling
- Implemented real-time 3D visualization with trajectory trails, velocity vectors, HUD telemetry, and CSV export for flight analysis

Porta Profile - Champaign, IL — Nov 2025
Fusion, PostgreSQL, Node.js, React.js

- Designed a credit-card sized stainless steel NFC multi-tool in Fusion that enables recruiters to instantly transfer contact information to candidates' phones at career fairs, eliminating reliance on paper business cards and lost resumes
- Developed a full-stack web platform (React.js, Node.js, PostgreSQL) with NFC/QR tap-to-share integration, awarded 2nd place at the Illinois Design Challenge

LEADERSHIP & INVOLVEMENT

Illinois Space Society (ISS) - Champaign, IL — Sept 2025-Dec 2025
Spaceshot, Structural Design & Analysis (SDA) Sub team Engineer

- Designed and analyzed Spaceshot structural components in Siemens NX to assess loads and guide design decisions
- Performed ANSYS FEA on fin assemblies to evaluate stress and deformation under launch and flight load